KRUSHANG PRAJAPATI

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Surat, Gujarat | DOB: 30 May 2006

Profile Overview

Third-year Electronics and Communication Engineering student focused on FPGA and digital design (Verilog HDL, Xilinx Vivado), with hands-on embedded systems and IoT experience. 1st place among 22 teams at FPGA Design Challenge 2.0 for a real-time image processing system on the Basys3 (Artix-7).

Interested in FPGA/VLSI design, hardware–software integration, edge AI and vision systems, and cybersecurity. Passionate about building practical systems that combine electronics, cloud platforms and intelligent automation. Seeking Frontend VLSI and FPGA Applications Engineering internships.

Education

B.Tech — Electronics and Communication Engineering (Pursuing) *2024 – 2028*
Sarvajanik College of Engineering & Technology, Sarvajanik University, Surat *SPI: 7.3/10 (Sem 4)*

HSC — Class XII, Science (PCM) | BhagwanMahavir School, Surat | CBSE *2024 · 64%*

SSC — Class X | B.S Patel Primary School, Bilimora | CBSE *2022 · 79%*

Skills

HDL & FPGA: Verilog HDL, Xilinx Vivado, Basys3 (Artix-7), FSM Design, Testbench Verification, Timing & Simulation
Embedded & IoT:ESP32, ATmega32 (AVR), Arduino, ESP32-CAM, Sensor Integration, Relay Modules, IoT System Design, IoT Dashboards
Electronics & Hardware: Analog & Digital Electronics, Op-Amps, Diodes, Transistors, Circuit Design & Prototyping
Circuit Simulation: LTspice, MATLAB Simulink, Vivado
Programming: C, Python, AVR Assembly, MATLAB
Cloud & Backend: Firebase Realtime Database, Firebase Authentication, Backend Integration
Systems: Linux (Arch Linux & Ubuntu), Git, GitHub, Command Line Tools
Machine Learning: Machine Learning Fundamentals, Motion Detection, Computer Vision, Offline LLM Assistant (Llama)
Web: React, TypeScript, Tailwind CSS, Node.js, HTML/CSS
Soft Skills: Critical Thinking, Leadership, Communication, Team Collaboration

Projects

FPGA-Based Image Processing System
Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA (Artix-7) | GitHub

- Designed and implemented a real-time image processing system on the Basys3 (Artix-7) FPGA in Verilog HDL, applying 8 image filters in hardware.
- Won 1st place among 22 teams at the FPGA Design Challenge 2.0, SCET, Surat.

Direct-Mapped Write-Back Cache Controller
Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA | GitHub

- Designed a direct-mapped, write-back, write-allocate cache controller in Verilog HDL: 5-state FSM (IDLE, COMPARE, WRITE_BACK, MEM_FETCH, UPDATE), 256 lines, 4-word line size.
- Verified with a testbench; added a Basys3 top-level (switches/LEDs) to demonstrate hits, misses and write-backs.

FPGA-Based Digital Design Implementations
Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA

- Implemented Sequence Detector (Moore & Mealy FSM), Sequence Generator, arithmetic units (Adder / Subtractor), Flip-Flops, MUX, and DSD modules in Verilog HDL — validated on FPGA.

XOR-Based Symmetric Encryption Engine
Academic Project | Verilog HDL | FPGA

- Implemented parameterised encryption / decryption hardware with configurable key width and key-locked decryption logic — validated on FPGA.

ESP32-CAM AI Motion Detection & Monitoring System
Academic Project | ESP 32 - CAM | FireBase | AI Motion Detection

- Developed a smart surveillance system using ESP32-CAM integrated with an AI-based motion detection model. Detected movement through camera input and sent alerts/data to a Firebase realtime dashboard for remote monitoring.

Offline AI Coding Assistant
Personal Project | Python | Llama | Local LLM | GitHub

- Built an offline, Llama-based assistant in Python that helps developers with code while keeping all data local and secure.

Mindscape — Mental Wellness & Focus Tracker
Personal Project | TypeScript | Android | GitHub

- Built an app for daily mood logging, intention setting and reflection, to support mental wellness and focus.

Python HTTP Honeypot
Personal Project | Python | Network Security | GitHub

- Built a lightweight HTTP honeypot that mimics a vulnerable web server and logs all incoming requests to detect unauthorized access, for security research.

Smart Classroom IoT Dashboard
Personal Project | TypeScript | IoT | GitHub

- Built a monitoring dashboard showing real-time temperature and occupancy data with device control.

Positions of Responsibility

Google Student Ambassador *Aug 2025 – Jan 2026*

- Explored Gemini AI capabilities; conducted sessions training college students on using AI tools for academic and professional workflows.

IEEE Student Member | T&P Cell Volunteer *2024 – 2025*

Achievements

- **1st Place — FPGA Design Challenge 2.0** (22 teams), SCET Surat: real-time image processing system on Basys3 (Artix-7)
- **3rd Place — Networking with Cisco Packet Tracer**, SCET Surat
- **Winner — Chess Tournament**, SCET Surat
- **Participant — International Press, SUMUN 2025**, Sarvajanik University
- **Volunteer — Cyber Workshop**, Sarvajanik University
- **Volunteer — SAW 3.0 Pre-Symposium 2025**, SCET